

Variance and local log-concavity of Poisson–binomial laws*

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Abstract

Let W be the number of successes in independent trials with unequal success probabilities, with probability mass function f and variance $V \geq 1$, and let D be the first index at which f decreases. The normalized Turán deficit $\delta_k = 1 - f_{k-1}f_{k+1}/f_k^2$, a measure of local log-concavity, is a bounded transform of the discrete curvature of $\log f$ at k ; for a normal density of variance V , that curvature is $1/V$. Quantitative log-concavity is used to derive concentration and log-Sobolev inequalities for discrete laws, but for Bernoulli sums the known lower bounds on δ_k depend on the number of summands or on the success probabilities. We prove $V\delta_D \geq 1/4$, and binomial laws show that no constant above $1/3$ is possible. With cubic inequalities of Hillion and Johnson, this gives $\delta_k \geq 1/(4V + |k - D|)$ on the whole support. Ultra-log-concavity alone gives no variance-scaled lower bound on δ_D . For the number of eigenvalues of a Haar unitary matrix in a half circle, our bound at D is of order $1/\log N$, against order $1/N$ from ultra-log-concavity. The proof of the main theorem is computer-assisted and is verified in the Lean proof assistant.

Keywords: Poisson–binomial law; log-concavity; Turán inequality; ultra-log-concavity; computer-assisted proof; formal verification.

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1 Introduction and main results

Let

$$W = \sum_{i=1}^n B_i, \quad B_i \sim \text{Bernoulli}(p_i),$$

where the summands are independent, so that W counts the successes in independent trials with unequal success probabilities. The law of W is Poisson–binomial; it arises in reliability analysis, survey sampling and finance, among other fields, and [14] is a recent survey.

Summands with $p_i = 0$ can be deleted, and summands with $p_i = 1$ only translate the law, which changes nothing below except by a shift of index (Section 2.1). We therefore assume throughout that every success probability satisfies $0 < p_i < 1$. Put

$$f_k = \mathbb{P}(W = k), \quad V = \text{Var}(W) = \sum_{i=1}^n p_i(1 - p_i),$$

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and call $(f_k)_{k=0}^n$ the probability mass function (pmf) of W . Extend the pmf by $f_k = 0$ for $k \notin \{0, \dots, n\}$. The probability-generating polynomial $\sum_{k=0}^n f_k x^k$ has only negative real zeros, so, by Newton's inequalities [12, equation (1)], the pmf is log-concave: $f_k^2 \geq f_{k-1}f_{k+1}$ for every k , an inequality of Turán type. For $0 \leq k \leq n$, we measure its slack relative to f_k^2 by the normalized Turán deficit

$$\delta_k := 1 - \frac{f_{k-1}f_{k+1}}{f_k^2}. \quad (1.1)$$

We have $0 \leq \delta_k \leq 1$ and $\delta_0 = \delta_n = 1$.

At an index with $f_{k-1}f_{k+1} > 0$,

$$\delta_k = 1 - e^{-C_k}, \quad C_k := 2 \log f_k - \log f_{k-1} - \log f_{k+1},$$

so δ_k is a bounded transform of the discrete curvature of $\log f$ at k . For a normal density of variance V , the corresponding curvature is $1/V$ at every point, which suggests $1/V$ as the natural scale for δ_k near the mode. Since δ_k depends on three adjacent masses and V on the whole law, we ask whether every Bernoulli sum satisfies $\delta_k \geq \kappa/V$ near the mode for a universal $\kappa > 0$, with no dependence on the number of summands or on the success probabilities.

Quantitative log-concavity has probabilistic consequences (see [13] for a review). Johnson [8] calls a pmf c -log-concave when $f_x/f_{x+1} - f_{x-1}/f_x \geq c$ for every x , a discrete curvature condition, and derives from it a modified log-Sobolev inequality and concentration of measure; Marsiglietti and Melbourne [10] derive concentration, moment and entropy bounds from log-concavity relative to a reference law. A constant c of Johnson's kind can never exceed $1/\mathbb{E}W$ [8, Lemma 5.3], so when many summands have p_i near 1 it lies far below the Gaussian scale $1/V$. At the first descent D defined below, if $D < n$, Johnson's quantity equals $(f_D/f_{D+1})\delta_D \geq \delta_D \geq 1/(4V)$ when $V \geq 1$, by Theorem 1.1.

The number of summands is the wrong scale for this question. Summands with p_i near 0 or 1 increase n but add little to V . The lower bounds on δ_k that we know of are not variance-scaled. They depend on n , such as the ultra-log-concavity bound recorded by Hillion and Johnson [6]; on the success probabilities, such as a bound of Johnson [8] and the bound that follows from Pitman's two-sided estimates of adjacent mass ratios through exponentially tilted laws [12, equation (20)]; or on k alone, such as the bound $\delta_k > 1/(k+1)$ that follows from the strict decrease of $(k+1)f_{k+1}/f_k$ proved by Dümbgen and Wellner [5, Proposition 2]. All four degenerate near the mode when summands with p_i near 0 and summands with p_i near 1 accumulate at fixed variance (Section 5.3 and Table 1). Darroch's theorem that every mode lies at distance less than one from the mean [4, Theorem 4] and the maximal-mass bounds of [1, 2] locate the mode or bound the largest mass, but by themselves give no lower bound on δ_k .

We state the bound at the first index where f decreases,

$$D := \min\{1 \leq k \leq n : f_k < f_{k-1}\}. \quad (1.2)$$

When $V \geq 1$, this set is nonempty and $2 \leq D \leq n$ (Section 2.1), and, by log-concavity, $c := D - 1$ is the rightmost modal index.

Theorem 1.1. *Let W be a finite sum of independent Bernoulli variables with success probabilities in $(0, 1)$, pmf f and variance $V \geq 1$, and let D be defined by (1.2). Then*

$$V\delta_D = V \left(1 - \frac{f_{D-1}f_{D+1}}{f_D^2} \right) \geq \frac{1}{4}. \quad (1.3)$$

Without some lower bound on V , no positive constant is possible. If $W \sim \text{Bernoulli}(p)$ with $0 < p < 1/2$, then $D = 1$, $\delta_D = 1$, and $V\delta_D = p(1-p) \rightarrow 0$ as $p \downarrow 0$.

Let κ_* denote the infimum of $V\delta_D$ over all finite Poisson–binomial laws with $V \geq 1$, where D and δ_D are defined from each law’s pmf as above. The theorem and the next proposition give

$$\frac{1}{4} \leq \kappa_* \leq \frac{1}{3}. \quad (1.4)$$

Proposition 1.2. *For every $n \geq 5$, there exists a binomial law of variance one whose pmf has first-descent index $D = 2$ and for which*

$$\delta_D = \frac{n+1}{3(n-1)}.$$

We do not know which, if either, endpoint of (1.4) equals κ_* ; in particular, we do not know whether the binomial laws of Proposition 1.2 are asymptotically extremal, which would give $\kappa_* = 1/3$. We are not aware of an earlier proof that $\kappa_* > 0$.

Cubic inequalities of Hillion and Johnson [6] for Bernoulli sums imply that $1/\delta_k$ changes by at most one per lattice step (Section 2), so the bound at D gives a bound at every point of the support.

Corollary 1.3. *Under the hypotheses of Theorem 1.1, for $0 \leq k \leq n$,*

$$\delta_k \geq \frac{1}{4V + |k - D|} \geq \frac{1}{4V + |k - \mathbb{E}W| + 2}, \quad (1.5)$$

and

$$V \left(1 - \frac{f_{D+1}}{f_D} \right) \geq \frac{1}{4}. \quad (1.6)$$

Theorem 1.1 and Corollary 1.3 also hold for differences of independent Bernoulli sums (Section 2.1). On a window $|k - \mathbb{E}W| \leq \alpha\sqrt{V}$, the bound (1.5) stays on the $1/V$ scale for fixed α . Whether a bound that varies over the support in this way yields a functional inequality, as Johnson’s c -log-concavity does, we leave open.

At the rightmost mode, δ_c is within constant factors of $1/V$; since $V \geq 1$, the next proposition gives $1/5 \leq V\delta_c < 2$.

Proposition 1.4. *If $V \geq 1$, then*

$$\frac{1}{4V+1} \leq \delta_c < \frac{2}{V}.$$

A pmf $(f_k)_{k=0}^n$ is ultra-log-concave of order n , written $\text{ULC}(n)$, if $f_k/\binom{n}{k}$ is log-concave in k (on such sequences, see [9]). Poisson–binomial pmfs are $\text{ULC}(n)$, and this gives

$$\delta_k \geq \frac{n+1}{(k+1)(n-k+1)}; \quad (1.7)$$

see [6, Definition 3.11 and equation (41)] and, for a recent treatment via relative log-concavity, [10, the discussion following Definition 1.3]. No variance-scaled lower bound on δ_D holds for ULC laws in general.

Proposition 1.5. *For $m \geq 1$, let $g = g^{(m)}$ be the pmf on $\{0, \dots, 2m\}$ with $g_{m+j} \propto \binom{2m}{m+j} 2^{-|j|}$. Then g is $\text{ULC}(2m)$ and strictly log-concave, its first-descent index is $m+1$, its deficit there is $(2m+1)/(m(m+2))$, and its variance V_m tends to 4. Hence $V_m\delta_{m+1} \rightarrow 0$.*

Remark 1.6. For $g^{(m)}$, $1/\delta_m \leq 4/3$ while $1/\delta_{m+1} \rightarrow \infty$, so $1/\delta$ changes by an unbounded amount in one step. By the reciprocal bound (2.5), this cannot happen for a Bernoulli sum.

The bound (1.7) also degenerates for Bernoulli sums themselves, and so do the other lower bounds on δ_k cited above. Table 1 shows this for a family of Bernoulli sums

Table 1: Lower bounds on δ_D , and δ_D itself, in the balanced family with $V = 1$ and $D = m + 1$, as $m \rightarrow \infty$.

Bound	Source	Value at D
Ultra-log-concavity (1.7)	[6]	$(2m + 1)/(m(m + 2)) \sim 2/m$
$\delta_k > 1/(k + 1)$	[5]	$1/(m + 2) \sim 1/m$
Odds-sum bound	[8]	$O(m^{-2})$
Combined tilt bounds	[12]	$\sim \sqrt{2}/m$
Theorem 1.1		$1/(4V) = 1/4$
δ_D itself		$\rightarrow 1 - I_0(1)I_2(1)/I_1(1)^2 \approx 0.46$

with $V = 1$ and $D = m + 1$: each cited bound at D is of order at most $1/m$, whereas Theorem 1.1 gives $1/4$ and δ_D itself tends to about 0.46 (Section 5.3).

A polynomial with nonnegative coefficients and only real zeros is, after normalization, the generating function of a Bernoulli sum [12, Proposition 1], so Theorem 1.1 and its consequences apply to the coefficient sequences of such polynomials, for example rows of the Stirling and Eulerian arrays [3], whenever the associated law has variance at least one.

Example 1.7. Determinantal point processes supply Bernoulli sums in which V and n differ in order. Let X_N count the eigenvalues of an $N \times N$ Haar unitary matrix in the open upper half of the unit circle, with N even. Meckes and Meckes [11, Proposition 3(1)] show that X_N has the law of a sum of N independent Bernoulli variables. Its variance is

$$V_N = \frac{\log N + \gamma + \log 2 + 1}{\pi^2} + O(N^{-2}),$$

where γ is Euler's constant, and $D = N/2 + 1$ (Section 5.4). Once $V_N \geq 1$, which holds for every even $N \geq 1998$, Theorem 1.1 gives $\delta_D \geq 1/(4V_N) \sim \pi^2/(4 \log N)$, whereas (1.7) gives only $(N + 1)/((N/2 + 2)N/2) \sim 4/N$. At $N = 10^4$ these lower bounds are about 0.215 and 0.0004.

The proof of Theorem 1.1 runs as follows. The reciprocal bound (2.5) shows that a small value of $\delta = \delta_D$ forces small values nearby: $\delta_{D \pm r} \leq \delta/(1 - r\delta)$ while $(r + 1)\delta < 1$. The masses at distance up to about $1/\delta$ from the mode are therefore bounded below by explicit multiples of the maximal mass M (Section 2). Summed in the pairwise variance identity, these bounds give $V \geq M^2 A(\delta)$ for an explicit function A , while a maximal-mass bound of Bobkov, Marsiglietti, and Melbourne gives $M^2 \geq (1 + 12V)^{-1}$. Together they give $V(1 + 12V) \geq A(\delta)$. Since $V(1 + 12V)$ increases with V and equals $(3 + \delta)/(4\delta^2)$ at $V = 1/(4\delta)$, the theorem reduces to the one-variable inequality $A(\delta) \geq (3 + \delta)/(4\delta^2)$ (Section 3).

Heuristically, the lower bounds on $f_{c \pm r}/M$ are about $e^{-r^2 \delta/2}$ when $r\delta$ is small, so the masses within distance about $\delta^{-1/2}$ of the mode are comparable to M , and $A(\delta)$ is of order δ^{-2} . With $V(1 + 12V) \geq A(\delta)$, this makes V of order at least $1/\delta$.

Section 4 proves the one-variable inequality, with a computer-assisted step in exact rational arithmetic, and Section 5 proves the other results of this section. The proof of Theorem 1.1, including the cubic inequalities of Hillion and Johnson, the strict log-concavity of the pmf, and the maximal-mass bound of Bobkov, Marsiglietti, and Melbourne, is also formalized in Lean (supplement). Several steps of the argument can lose a constant, so it does not determine κ_* .

2 The recurrence for normalized Turán deficits

We show that a small deficit at D forces small deficits nearby and deduce lower bounds on the masses near the mode.

2.1 Standing reductions

Let W be a finite Bernoulli sum with arbitrary success probabilities $p_i \in [0, 1]$. Put $I = \{i : 0 < p_i < 1\}$, $n_1 = \#\{i : p_i = 1\}$, and $W^\circ = \sum_{i \in I} B_i$. Then $W = n_1 + W^\circ$ and $\text{Var}(W^\circ) = \text{Var}(W)$. Let $h = (h_k)$ be the pmf of W° , where $h_k = \mathbb{P}(W^\circ = k)$. Then $\mathbb{P}(W = n_1 + k) = h_k$; hence translation shifts the first-descent index by n_1 and preserves the corresponding adjacent mass ratios and normalized Turán deficits wherever defined. If $\text{Var}(W) \geq 1$, then I is nonempty, so we may replace W by W° , relabel $|I|$ as n , and assume $0 < p_i < 1$ for every i , as in Section 1.

Under this assumption and $V \geq 1$, the set in (1.2) is nonempty. Indeed,

$$\frac{f_n}{f_{n-1}} = \left(\sum_{i=1}^n \frac{1-p_i}{p_i} \right)^{-1}.$$

If $f_n \geq f_{n-1}$, then the sum on the right is at most one, whereas

$$V = \sum_{i=1}^n p_i^2 \frac{1-p_i}{p_i} < \sum_{i=1}^n \frac{1-p_i}{p_i} \leq 1,$$

a contradiction. If $f_1 < f_0$, then by log-concavity 0 is a mode, so $\mathbb{E}W < 1$ by Darroch's theorem [4, Theorem 4] and $V \leq \sum_i p_i = \mathbb{E}W < 1$. Hence $D \geq 2$.

If $W' = \sum_{j=1}^m B'_j$ is a Bernoulli sum independent of W , then $W - W' + m = W + \sum_j (1 - B'_j)$ is a Bernoulli sum with the variance of $W - W'$, and its pmf is that of $W - W'$ translated by m . Hence Theorem 1.1 and Corollary 1.3 apply to $W - W'$, with D the least k in the support of $W - W'$ such that $\mathbb{P}(W - W' = k) < \mathbb{P}(W - W' = k - 1)$.

2.2 The recurrence

For $1 \leq k \leq n$, put

$$q_k = \frac{f_k}{f_{k-1}}.$$

Log-concavity says that (q_k) is nonincreasing, and for $1 \leq k \leq n - 1$,

$$\delta_k = 1 - \frac{q_{k+1}}{q_k}. \quad (2.1)$$

For the pmf of any finite sum of independent Bernoulli variables, Hillion and Johnson's Theorem A.2 and Corollary A.3 state, for every $k \in \mathbb{Z}$, the cubic inequalities [6, equations (78)–(79)]

$$(f_{k-1}^2 - f_{k-2}f_k)f_{k+1} \leq f_{k-1}(f_k^2 - f_{k-1}f_{k+1}), \quad (2.2)$$

$$(f_{k+1}^2 - f_k f_{k+2})f_{k-1} \leq f_{k+1}(f_k^2 - f_{k-1}f_{k+1}). \quad (2.3)$$

Hillion and Johnson extend the pmf by zero outside its support. Since

$$f_{k-1}^2 - f_{k-2}f_k = f_{k-1}^2 \delta_{k-1},$$

$$f_k^2 - f_{k-1}f_{k+1} = f_k^2 \delta_k,$$

$$f_{k+1}^2 - f_k f_{k+2} = f_{k+1}^2 \delta_{k+1},$$

dividing (2.2) by $f_{k-1}f_k^2$ and (2.3) by $f_{k+1}f_k^2$ gives, for $1 \leq k \leq n - 1$, the recurrence

$$\delta_{k-1}(1 - \delta_k) \leq \delta_k, \quad \delta_{k+1}(1 - \delta_k) \leq \delta_k. \quad (2.4)$$

Every δ_k is positive by (1.7), so dividing the second inequality by $\delta_k \delta_{k+1}$ gives $1/\delta_{k+1} \geq 1/\delta_k - 1$, and the first gives the same with $k-1$ in place of $k+1$. The two inequalities not given by (2.4), $1/\delta_1 - 1/\delta_0 \geq -1$ and $1/\delta_{n-1} - 1/\delta_n \geq -1$, hold because $1/\delta_0 = 1/\delta_n = 1 \leq 1/\delta_k$ for every k . Hence, for $0 \leq j \leq n-1$,

$$\left| \frac{1}{\delta_{j+1}} - \frac{1}{\delta_j} \right| \leq 1. \quad (2.5)$$

Set $\delta = \delta_D$. If $\delta \geq 1/4$, Theorem 1.1 follows from $V \geq 1$, so it remains to consider

$$0 < \delta < \frac{1}{4}. \quad (2.6)$$

Define

$$K = \max\{r \in \mathbb{Z}_{\geq 1} : (r+1)\delta < 1\}, \quad a = \frac{1-2\delta}{1-\delta}. \quad (2.7)$$

By (2.6), $K \geq 3$.

Lemma 2.1. Assume (2.6). For $0 \leq r \leq K$ and each sign, $D \pm r \in \{1, \dots, n-1\}$ and

$$\delta_{D \pm r} \leq \frac{\delta}{1-r\delta} < 1. \quad (2.8)$$

Proof. We induct on r . For $r = 0$, we have $D \in \{1, \dots, n-1\}$ because $2 \leq D \leq n$ and $\delta_n = 1 > \delta$, and (2.8) holds with equality. Let $1 \leq r \leq K$ and suppose $D \pm (r-1) \in \{1, \dots, n-1\}$. Then $D \pm r \in \{0, \dots, n\}$, and r applications of (2.5) give $1/\delta_{D \pm r} \geq 1/\delta - r > 1$, because $(r+1)\delta < 1$. So $D \pm r$ is neither 0 nor n , where the deficit equals one, and inverting gives (2.8). $\square$

By Lemma 2.1, $D - K \geq 1$ and $D + K \leq n-1$, so every index $c \pm r$ with $0 \leq r \leq K$ lies in the support. Write $M = f_c = \max_k f_k$ for the maximal mass. Since $D \geq K+1 \geq 2$, the ratio q_{D-1} is defined, and $q_{D-1} \geq 1$ by the definition of D . The lemma with $r = 1$ gives $\delta_{D-1} \leq \delta/(1-\delta)$, so

$$q_D = q_{D-1}(1-\delta_{D-1}) \geq a.$$

For $0 \leq j \leq K-1$, (2.8) gives

$$\begin{aligned} q_{D+j} &= q_D \prod_{s=0}^{j-1} (1-\delta_{D+s}) \\ &\geq a \prod_{s=0}^{j-1} \frac{1-(s+1)\delta}{1-s\delta} = a(1-j\delta). \end{aligned}$$

For $1 \leq r \leq K$, we obtain the lower bound

$$R_r := a^r \prod_{j=1}^{r-1} (1-j\delta), \quad \frac{f_{c+r}}{M} = \prod_{j=0}^{r-1} q_{D+j} \geq R_r. \quad (2.9)$$

On the left, let $1 \leq j \leq K$. Then

$$q_D = q_{D-j} \prod_{s=1}^j (1-\delta_{D-s}) < 1,$$

and hence

$$\begin{aligned} \frac{1}{q_{D-j}} &> \prod_{s=1}^j (1-\delta_{D-s}) \\ &\geq \prod_{s=1}^j \frac{1-(s+1)\delta}{1-s\delta} = \frac{1-(j+1)\delta}{1-\delta}. \end{aligned}$$

For $1 \leq r \leq K$, we obtain the lower bound

$$L_r := (1 - \delta)^{-r} \prod_{j=2}^{r+1} (1 - j\delta), \quad \frac{f_{c-r}}{M} = \prod_{j=1}^r \frac{1}{q_{D-j}} \geq L_r. \quad (2.10)$$

3 Variance bounds and scalar reduction

We turn these mass bounds into a lower bound on V and reduce Theorem 1.1 to a one-variable inequality. Define auxiliary weights by $w_0 = 1$, $w_r = R_r$, and $w_{-r} = L_r$ for $1 \leq r \leq K$, and put

$$A(\delta) = \frac{1}{2} \sum_{i,j=-K}^K w_i w_j (i - j)^2. \quad (3.1)$$

Restricting the pairwise variance identity to indices $c - K, \dots, c + K$, which lie in the support, and using $f_{c+i} \geq M w_i$ gives

$$V = \frac{1}{2} \sum_{i,j=0}^n f_i f_j (i - j)^2 \geq M^2 A(\delta). \quad (3.2)$$

We also use the maximal-mass bound of Bobkov, Marsiglietti, and Melbourne for integer-valued laws [2, Corollary 3.2]:

$$V \geq \frac{M^{-2} - 1}{12}. \quad (3.3)$$

For completeness, here is a short proof. Let U be independent of W and uniform on $[-1/2, 1/2]$. Then $W + U$ has a density φ bounded by M , and $\text{Var}(W + U) = V + 1/12$. Put $x_0 = \mathbb{E}(W + U)$ and $\psi = M \mathbf{1}\{|x - x_0| \leq 1/(2M)\}$. Both φ and ψ integrate to one, and $(\varphi - \psi)(x)((x - x_0)^2 - (2M)^{-2}) \geq 0$ for every x . Hence

$$V + \frac{1}{12} = \int_{\mathbb{R}} (x - x_0)^2 \varphi(x) dx \geq \int_{\mathbb{R}} (x - x_0)^2 \psi(x) dx = \frac{1}{12M^2}.$$

It remains to prove one scalar estimate.

Proposition 3.1. *For $0 < \delta < 1/4$, the quantity in (3.1) satisfies*

$$A(\delta) \geq \frac{3 + \delta}{4\delta^2}. \quad (3.4)$$

Proof of Theorem 1.1 from Proposition 3.1. By the reduction preceding (2.6), we may assume $0 < \delta < 1/4$. Suppose that $V < 1/(4\delta)$. Inequality (3.3) then gives

$$M^2 > \frac{\delta}{3 + \delta}.$$

On the other hand, (3.2) and (3.4) give

$$M^2 < \frac{1}{4\delta A(\delta)} \leq \frac{\delta}{3 + \delta},$$

a contradiction. □

4 The scalar inequality

Set

$$H = \frac{1 - \delta}{\delta} = \frac{1}{\delta} - 1 > 3. \quad (4.1)$$

Then $K = \max\{r \in \mathbb{Z}_{\geq 1} : r < H\}$, and the bounds L_r of (2.10) become

$$L_r = b_r := \prod_{s=1}^r \left(1 - \frac{s}{H}\right), \quad (4.2)$$

which are positive because $r \leq K < H$. With K fixed, the form (3.1) is a quadratic polynomial in the weights with nonnegative coefficients, so its value does not decrease when nonnegative weights are increased componentwise. Under (4.1), the right-hand side of (3.4) is

$$Q(H) = \frac{(3H+4)(H+1)}{4}. \quad (4.3)$$

Section 4.1 treats $3 < H \leq 16$ with the weights as they stand, and Section 4.2 symmetrizes them to treat $H \geq 16$.

4.1 The range $3 < H \leq 16$

For an integer $K \geq 3$ and $0 < \delta \leq 1/(K+1)$, let $A_K(\delta)$ denote the right-hand side of (3.1) with K held fixed, so that $A(\delta) = A_K(\delta)$ when K is given by (2.7). On this range every weight is nonnegative and nonincreasing in δ : $a = 1 - \delta/(1 - \delta)$ and the factors $1 - j\delta$, $j < K$, are nonnegative and decreasing, so R_r is; and $L_r = \prod_{s=1}^r (1 - s\delta/(1 - \delta))$ by (4.2), with factors in $[0, 1]$, because $(s+1)\delta \leq 1$, and decreasing. Since A_K does not decrease when nonnegative weights increase, $A_K(\delta)$ is nonincreasing in δ , and so is $(3 + \delta)/(4\delta^2)$. Hence, if $0 < \delta_- < \delta_+ \leq 1/(K+1)$ and

$$\frac{3 + \delta_-}{4\delta_-^2} \leq A_K(\delta_+), \quad (4.4)$$

then $A_K(\delta) \geq A_K(\delta_+) \geq (3 + \delta_-)/(4\delta_-^2) \geq (3 + \delta)/(4\delta^2)$ for every $\delta \in [\delta_-, \delta_+]$.

For $m = 4, \dots, 15$, the range $m < H \leq m+1$ is $1/(m+2) \leq \delta < 1/(m+1)$, where $K = m$. We apply (4.4) with $K = m$, $\delta_- = 1/(m+2)$ and $\delta_+ = 1/(m+1)$. The ratios of its right-hand side to its left-hand side, rounded down, are 1.072, 1.439, 1.763, 2.050, 2.304, 2.531, 2.735, 2.920, 3.089, 3.243, 3.385 and 3.516 for $m = 4, \dots, 15$. For $3 < H \leq 4$, that is $1/5 \leq \delta < 1/4$, we have $K = 3$; we split $[1/5, 1/4]$ into 20 intervals of length $1/400$ and apply (4.4) on each, and the smallest ratio, on the interval ending at $1/4$, is 1.0078 rounded down. A program in the supplement checks these 32 inequalities in exact rational arithmetic, and the Lean formalization described there checks them in the Lean kernel. This proves (3.4) for $3 < H \leq 16$, that is, for $1/17 \leq \delta < 1/4$.

4.2 The range $H \geq 16$

For $1 \leq r \leq K$, the bounds R_r of (2.9) satisfy $R_r \geq b_r$. Indeed, if $C_r = R_r/L_r$, then $C_1 = 1$ and, for $1 \leq r < K$ (so that $H - r - 1 > 0$),

$$\frac{C_{r+1}}{C_r} = \frac{(H-1)(H+1-r)}{(H+1)(H-r-1)} = 1 + \frac{2r}{(H+1)(H-r-1)} \geq 1. \quad (4.5)$$

Replacing every R_r by b_r therefore does not increase A ; denote the resulting value by A_{sym} . For the symmetric weights $w_0 = 1$ and $w_{\pm r} = b_r$, put

$$S = 1 + 2 \sum_{r=1}^K b_r, \quad T = 2 \sum_{r=1}^K r^2 b_r. \quad (4.6)$$

The pairwise identity gives

$$A_{\text{sym}} = \left(\sum_{r=-K}^K w_r \right) \left(\sum_{r=-K}^K r^2 w_r \right) - \left(\sum_{r=-K}^K r w_r \right)^2 = ST,$$

because the final sum vanishes by symmetry. Hence $A(\delta) \geq A_{\text{sym}} = ST$. This bound is too weak near $H = 3$ ($ST < Q(H)$ at $H = 7/2$, for example), so we use it only for $H \geq 16$, where it remains to prove $ST \geq Q(H)$.

If $G(t) = \sum_{j=0}^d a_j t^j$ has degree d , its coefficients in the Bernstein basis on $[0, 1]$ are

$$\beta_i = \sum_{j=0}^i a_j \frac{\binom{i}{j}}{\binom{d}{j}}, \quad G(t) = \sum_{i=0}^d \beta_i \binom{d}{i} t^i (1-t)^{d-i}, \quad (4.7)$$

and $G \geq 0$ on $[0, 1]$ when every β_i is nonnegative, since the basis polynomials are nonnegative there.

We cover $H \geq 16$ by the intervals between consecutive triangular numbers: choose an integer $J \geq 5$ such that

$$\frac{J(J+1)}{2} \leq H \leq \frac{(J+1)(J+2)}{2}. \quad (4.8)$$

Either choice of J is valid at a shared endpoint. The first interval ($J = 5$) is checked directly, and the rest ($J \geq 6$) uniformly in J . If $J \geq 5$, then $J < J(J+1)/2 \leq H$, so $J < K$. For $1 \leq r \leq J$, the inequality $\prod_s (1 - x_s) \geq 1 - \sum_s x_s$ for $x_s \in [0, 1]$ gives

$$b_r \geq \lambda_r := 1 - \frac{r(r+1)}{2H} \geq 0, \quad (4.9)$$

where $\lambda_r \geq 0$ because $r(r+1)/2 \leq J(J+1)/2 \leq H$; this is why the cells begin at triangular numbers. Put

$$\tilde{S}_J = 1 + 2 \sum_{r=1}^J \lambda_r = 2J + 1 - \frac{J(J+1)(J+2)}{3H}, \quad (4.10)$$

$$\tilde{T}_J = 2 \sum_{r=1}^J r^2 \lambda_r = \frac{J(J+1)(2J+1)}{3} - \frac{\sigma_3 + \sigma_4}{H}, \quad (4.11)$$

where

$$\sigma_3 = \sum_{r=1}^J r^3 = \frac{J^2(J+1)^2}{4}, \quad \sigma_4 = \sum_{r=1}^J r^4 = \frac{J(J+1)(2J+1)(3J^2+3J-1)}{30}.$$

Since $J < K$, all omitted weights are nonnegative and $b_r \geq \lambda_r$ for $r \leq J$. Hence $S \geq \tilde{S}_J$, $T \geq \tilde{T}_J$, and $A(\delta) \geq ST \geq \tilde{S}_J \tilde{T}_J$. It remains to prove $\tilde{S}_J \tilde{T}_J \geq Q(H)$.

Let $N_J(H) = H^2(\tilde{S}_J \tilde{T}_J - Q(H))$. Expanding (4.10), (4.11), and (4.3) gives

$$\begin{aligned} N_J(H) = & -\frac{3}{4}H^4 - \frac{7}{4}H^3 + \left(\frac{J(J+1)(2J+1)^2}{3} - 1 \right) H^2 \\ & - \left((2J+1)(\sigma_3 + \sigma_4) + \frac{J^2(J+1)^2(J+2)(2J+1)}{9} \right) H \\ & + \frac{J(J+1)(J+2)}{3}(\sigma_3 + \sigma_4). \end{aligned}$$

For $J = 5$, only $16 \leq H \leq 21$ is needed. With $H = 16 + 5t$ and $0 \leq t \leq 1$, the degree-four Bernstein coefficients of $N_5(16 + 5t)$ in t are

$$2360, \quad 7500, \quad \frac{25055}{2}, \quad \frac{254205}{16}, \quad \frac{31115}{2}.$$

For $J \geq 6$, write

$$H = \frac{J(J+1)}{2} + (J+1)t, \quad J = u + 6, \quad 0 \leq t \leq 1.$$

The degree-four Bernstein coefficients in t of $N_J(J(J+1)/2 + (J+1)t)$ are $\beta_i = \mu_i \pi_i(u)/2880$ for $i = 0, \dots, 4$, where

$$\begin{aligned}\mu_0 &= J^2(J+1)^2, & \mu_1 &= J(J+1)^2, & \mu_2 &= (J+1)^2, \\ \mu_3 &= (J+1)^2(J+2), & \mu_4 &= (J+1)^2(J+2)^2,\end{aligned}$$

and

$$\begin{aligned}\pi_0(u) &= 121u^4 + 2474u^3 + 17431u^2 + 46014u + 25400, \\ \pi_1(u) &= 121u^5 + 3442u^4 + 37967u^3 + 199405u^2 + 480475u + 384510, \\ \pi_2(u) &= 121u^6 + 4410u^5 + 65863u^4 + 512764u^3 + 2172926u^2 + 4668316u + 3829440, \\ \pi_3(u) &= 121u^5 + 3684u^4 + 43735u^3 + 249961u^2 + 671859u + 644400, \\ \pi_4(u) &= 121u^4 + 2958u^3 + 25579u^2 + 88782u + 91440.\end{aligned}$$

Each μ_i is positive, and each π_i has positive coefficients. Thus $N_J(H) > 0$ and $\tilde{S}_J \tilde{T}_J > Q(H)$ for $H \geq 16$, which with Section 4.1 proves Proposition 3.1. A short SymPy program in the supplement checks every expansion in this subsection in exact arithmetic.

5 Proofs of the results of Section 1

5.1 The constant and the support

Proof of Proposition 1.2. Set $p_n = (1 - \sqrt{1 - 4/n})/2$ and $W_n \sim \text{Bin}(n, p_n)$. Then $np_n(1 - p_n) = 1$. Write $f = (f_k)$ for the pmf of W_n and put $q_k = f_k/f_{k-1}$. Then

$$q_k = \frac{n - k + 1}{k} \frac{p_n}{1 - p_n}.$$

Since $p_n(1 - p_n) = 1/n$, we have $p_n > 1/n$, so $q_1 = np_n/(1 - p_n) > 1$. Both p_n , the smaller root of $x(1 - x) = 1/n$, and $2/(n+1)$ lie in $(0, 1/2)$, where $x \mapsto x(1 - x)$ is increasing. For $n \geq 5$ we have $n^2 - 4n - 1 > 0$, which is equivalent to

$$\frac{2}{n+1} \left(1 - \frac{2}{n+1}\right) = \frac{2(n-1)}{(n+1)^2} > \frac{1}{n},$$

so $p_n < 2/(n+1)$. This is equivalent to $q_2 < 1$, so $D = 2$. Finally,

$$\delta_2 = 1 - \frac{q_3}{q_2} = 1 - \frac{2(n-2)}{3(n-1)} = \frac{n+1}{3(n-1)} \rightarrow \frac{1}{3}.$$

□

Proof of Corollary 1.3. By the reciprocal bound (2.5) and Theorem 1.1, $1/\delta_k \leq 1/\delta_D + |k - D| \leq 4V + |k - D|$. Darroch's theorem [4, Theorem 4] gives $|c - \mathbb{E}W| < 1$, so $|D - \mathbb{E}W| < 2$, which gives the second inequality. For (1.6), write $\eta_- = f_D/f_{D-1} < 1$ and $\eta_+ = f_{D+1}/f_D$; then $1 - \eta_+ \geq 1 - \eta_+/\eta_- = \delta_D$. □

5.2 The rightmost mode

Proof of Proposition 1.4. The lower bound is (1.5) with $k = c$. For the upper bound, the idea is that exponential tilting moves the mean at a rate equal to the tilted variance: the tilts at which $c - 1, c$ and then $c, c + 1$ are the modes differ by $-\log(1 - \delta_c)$, and by Darroch's theorem the mean moves by less than 2 between them. For $t \in \mathbb{R}$, let $f_k^{(t)} = e^{tk} f_k / \sum_j e^{tj} f_j$, the pmf of the Bernoulli sum with parameters $p_i e^t / (1 - p_i + p_i e^t)$, and write $\mu(t)$ and $V(t)$ for its mean and variance, so $\mu'(t) = V(t)$. The tilted variance of a

summand is $p_i(1-p_i)e^t/(1-p_i+p_ie^t)^2 \geq e^{-|t|}p_i(1-p_i)$, so $V(t) \geq e^{-|t|}V$. Since $2 \leq D \leq n$, the ratios $q_c = f_c/f_{c-1} \geq 1$ and $q_{c+1} = f_{c+1}/f_c < 1$ are defined; put $t_- = -\log q_c \leq 0$ and $t_+ = -\log q_{c+1} > 0$. The pmf is strictly log-concave (by (1.7)), so $f^{(t_-)}$ has exactly the two modes $c-1$ and c , and $f^{(t_+)}$ exactly the two modes c and $c+1$. By Darroch's theorem [4, Theorem 4], applied to these tilted laws (which are again Poisson-binomial), $\mu(t_-) \in (c-1, c)$ and $\mu(t_+) \in (c, c+1)$. Hence

$$2 > \mu(t_+) - \mu(t_-) = \int_{t_-}^{t_+} V(t) dt \geq V(2 - e^{t_-} - e^{-t_+}) = V\left(2 - \frac{1}{q_c} - q_{c+1}\right) \geq V\left(1 - \frac{q_{c+1}}{q_c}\right),$$

where the last step uses $(1 - 1/q_c)(1 - q_{c+1}) \geq 0$, and $1 - q_{c+1}/q_c = \delta_c$. $\square$

5.3 Ultra-log-concavity and the balanced family

Proof of Proposition 1.5. The ratio $g_{m+j}/\binom{2m}{m+j} \propto 2^{-|j|}$ is log-concave, and the product of the strictly log-concave binomial row with it is strictly log-concave. The pmf is symmetric about m , so m is its unique mode and $m+1$ its first descent. There the geometric factors cancel: $\delta_{m+1} = 1 - \binom{2m}{m}\binom{2m}{m+2}\binom{2m}{m+1}^{-2} = (2m+1)/(m(m+2))$. Finally, $\binom{2m}{m+j}/\binom{2m}{m}$ increases to 1 as $m \rightarrow \infty$, for each fixed j . The mean is m by symmetry, so by dominated convergence with the envelope $(1+j^2)2^{-|j|}$ the variance converges to that of the law on $\mathbb{Z}$ proportional to $2^{-|j|}$, which is 4. $\square$

For $m \geq 2$, take m success probabilities equal to ε_m and m equal to $1 - \varepsilon_m$, where $\varepsilon_m \leq 1/2$ solves $2m\varepsilon_m(1 - \varepsilon_m) = 1$. This balanced family has $V = 1$, and its pmf is symmetric about m and strictly log-concave, so $D = m+1$. Moreover $W - m$ converges in law to the difference of two independent Poisson(1/2) variables, whose pmf is $e^{-1}I_k(1)$ with I_k the modified Bessel function, which gives the last row of Table 1. The first two rows are (1.7) and the bound $\delta_k > 1/(k+1)$ at $k = D$ and $n = 2m$.

Johnson's bound [8, Definition 1.2, Lemma 5.1, and Example 5.2(2)] reads $\delta_k \geq (f_{k+1}/f_k)(\sum_i p_i/(1-p_i))^{-1}$. Here the odds sum is at least $m(1 - \varepsilon_m)/\varepsilon_m \sim 2m^2$, so the bound is $O(m^{-2})$.

Pitman's bounds [12, equation (20)] on adjacent mass ratios are stated through the exponentially tilted laws $f^{(t)}$ of the proof of Proposition 1.4. Let $\theta(x) = e^t$, where t is the tilt at which $f^{(t)}$ has mean x . Combining Pitman's bounds at $k-1$ and k gives

$$\delta_k \geq 1 - \frac{\theta(k - 1/(n - k + 2))}{\theta(k + 1/(k + 2))}. \quad (5.1)$$

In the balanced family the tilted mean is $m + \sinh t/(1 + (\cosh t - 1)/m)$. At $k = D$ the two arguments of θ differ by $2/m + O(m^{-2})$, near the point $t = \operatorname{arsinh} 1$, where the tilted mean is $m + 1 + O(1/m)$ and has slope $\sqrt{2} + O(1/m)$. Since $\log \theta$ is the inverse of the tilted mean, the two values of $\log \theta$ differ by $\sqrt{2}/m + O(m^{-2})$, and so does the right-hand side of (5.1).

5.4 The Haar unitary example

By the theorem of Hough, Krishnapur, Peres, and Virág on which the result of Meckes and Meckes rests [7, Theorem 7], the parameters are the eigenvalues of the kernel of the eigenvalue process, a projection of rank N , compressed to the half circle. These are the nonzero eigenvalues of the Gram matrix Γ of the functions $e^{ij\phi}$, $0 \leq j < N$, on the half circle, with entries $\Gamma_{jk} = (2\pi)^{-1} \int_0^\pi e^{i(j-k)\phi} d\phi$. Both Γ and $I - \Gamma$ are Gram matrices of linearly independent functions on a half circle, so all N parameters lie in $(0, 1)$. Hence V_N , the sum of $p(1-p)$ over the eigenvalues p of Γ , is $\operatorname{tr} \Gamma - \operatorname{tr} \Gamma^2$, and since

$|\Gamma_{jk}|^2 = \pi^{-2}(j-k)^{-2}$ for odd $j-k$ and $\Gamma_{jk} = 0$ for even $j-k \neq 0$,

$$V_N = \frac{N}{4} - \frac{2}{\pi^2} \sum_{\substack{1 \leq d \leq N \\ d \text{ odd}}} \frac{N-d}{d^2} = \frac{2}{\pi^2} \sum_{d \text{ odd}} \frac{\min(d, N)}{d^2} = \frac{\log N + \gamma + \log 2 + 1}{\pi^2} + O(N^{-2}),$$

using $\sum_{d \text{ odd}} d^{-2} = \pi^2/8$; here γ is Euler's constant. Rotation by π shows that X_N and $N - X_N$ have the same law, so $c = N/2$ and $D = N/2 + 1$. The middle form shows that V_N increases with N . A program in the supplement evaluates the first form with rational bounds on π and finds $V_{1996} < 1 < V_{1998}$, so among even N the variance first reaches 1 at $N = 1998$.

Supplementary Material

Code and exact certificate data. The archive is at <https://doi.org/10.5281/zenodo.23129480>, with a manifest giving the software versions and the commands to rerun every check.

Computations the proof needs. The proof of Proposition 3.1 consists of the arguments of Section 4 together with two exact computations: a program using only exact rational arithmetic from the Python standard library checks the 32 inequalities of Section 4.1, and a SymPy program checks the expansions for $H \geq 16$. The deduction of Theorem 1.1 from Proposition 3.1 (Sections 2 and 3) is given in full in the text and uses no computation. A third program evaluates V_{1996} and V_{1998} in Example 1.7 with rational bounds on π .

Additional check. For $3 < H \leq 16$ the archive also contains a certificate by Bernstein expansions of $A(\delta) - Q(H)$ on $3 < H \leq 4$ and of $ST - Q(H)$ on $4 \leq H \leq 16$: 275 positive rational coefficients, generated by a SymPy program and recomputed by an independently written checker.

Formalization. The archive contains formalizations in the Lean 4 proof assistant with the Mathlib library, each depending only on Lean's three standard axioms. The first proves Proposition 3.1 by the route of Section 4. The second, which builds on it, proves Theorem 1.1 for every Bernoulli sum with $0 < p_i < 1$, starting from the probability-generating polynomial; it includes the cubic inequalities (2.2)–(2.3), the strict log-concavity of the pmf, and the maximal-mass bound (3.3), for which it gives a discrete proof. An earlier, conditional formalization covers Lemma 2.1 and the deduction of (1.6) from (1.3). The other results of Section 1 are not formalized.

References

- [1] Jean-Bernard Baillon, Roberto Cominetti, and José Vaisman, *A sharp uniform bound for the distribution of sums of Bernoulli trials*, *Combinatorics, Probability and Computing* **25** (2016), no. 3, 352–361. MR3482659
- [2] Sergey G. Bobkov, Arnaud Marsiglietti, and James Melbourne, *Concentration functions and entropy bounds for discrete log-concave distributions*, *Combinatorics, Probability and Computing* **31** (2022), no. 1, 54–72. MR4356456
- [3] Petter Brändén, *Unimodality, log-concavity, real-rootedness and beyond*, *Handbook of Enumerative Combinatorics, Discrete Mathematics and its Applications*, CRC Press, Boca Raton, FL, 2015, pp. 437–483. MR3409348
- [4] J. N. Darroch, *On the distribution of the number of successes in independent trials*, *The Annals of Mathematical Statistics* **35** (1964), no. 3, 1317–1321. MR164359
- [5] Lutz Dümbgen and Jon A. Wellner, *The density ratio of Poisson binomial versus Poisson distributions*, *Statistics & Probability Letters* **165** (2020), 108862. MR4118941

- [6] Erwan Hillion and Oliver Johnson, *Discrete versions of the transport equation and the Shepp–Olkin conjecture*, The Annals of Probability **44** (2016), no. 1, 276–306. MR3456338
- [7] J. Ben Hough, Manjunath Krishnapur, Yuval Peres, and Bálint Virág, *Determinantal processes and independence*, Probability Surveys **3** (2006), 206–229. MR2216966
- [8] Oliver Johnson, *A discrete log-Sobolev inequality under a Bakry–Émery type condition*, Annales de l’Institut Henri Poincaré, Probabilités et Statistiques **53** (2017), no. 4, 1952–1970. MR3729642
- [9] Thomas M. Liggett, *Ultra logconcave sequences and negative dependence*, Journal of Combinatorial Theory, Series A **79** (1997), no. 2, 315–325. MR1462561
- [10] Arnaud Marsiglietti and James Melbourne, *Concentration inequalities for log-concave sequences*, International Mathematics Research Notices **2026** (2026), no. 4, rnag023. MR5033003
- [11] Elizabeth S. Meckes and Mark W. Meckes, *A sharp rate of convergence for the empirical spectral measure of a random unitary matrix*, Zapiski Nauchnykh Seminarov POMI **457** (2017), 276–285. MR3723586
- [12] Jim Pitman, *Probabilistic bounds on the coefficients of polynomials with only real zeros*, Journal of Combinatorial Theory, Series A **77** (1997), no. 2, 279–303. MR1429082
- [13] Adrien Saumard and Jon A. Wellner, *Log-concavity and strong log-concavity: a review*, Statistics Surveys **8** (2014), 45–114. MR3290441
- [14] Wenpin Tang and Fengmin Tang, *The Poisson binomial distribution—old & new*, Statistical Science **38** (2023), no. 1, 108–119. MR4534645

AI use and author responsibility. OpenAI’s GPT-5.6 model, accessed through Codex, supplied substantial proof-search assistance for Theorem 1.1. An AI-generated review of an earlier draft (ChatGPT, OpenAI) proposed the framing in terms of variance and local log-concavity, Corollary 1.3, Proposition 1.4 with its proof, Proposition 1.5, and Example 1.7; each was checked, by exact computation on test laws and by independent model-based proof review, before it was adopted. A second ChatGPT review pointed out the bound (5.1) and independently reconstructed the Bernstein-expansion computations, with exact agreement. Aristotle (Harmonic), an automated theorem prover, produced the Lean proofs in the formalizations described in the supplement, and its proof of Proposition 3.1 supplied the monotonicity argument of Section 4.1. Further AI-generated reviews (ChatGPT, Elicit, and Codex with the model GPT-6.1-sol) checked the computations and the Lean statements, found a boundary case, suggested background references, and prompted corrections and the reorganization of Section 1. Claude Opus 5.5 (Anthropic), through Claude Code, assisted with revision, proof checking, exact-verification code, and source checking, and other sessions with Claude, ChatGPT and Codex (both OpenAI), and Gemini (Google) assisted with computational exploration, proof critique, formalization support, literature searching, and drafting. The archive’s provenance table lists each contribution with its date and model label. The author remains responsible for the mathematics, sources, wording, and released files.